

Ahmed Elfiky

AI ENGINEER

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PROFILE

AI Engineer with production experience building agentic systems, LLM applications, and document intelligence. Designs multi-agent workflows on Azure AI Foundry around function calling, tool use, and MCP, and evaluates their output with LLM-as-a-judge before release. Outside of work, fine-tunes and serves open-source models with QLoRA and vLLM, and ships optimized inference under tight compute budgets.

WORK EXPERIENCE

AI Engineer | Palm Hills Developments Oct 2024 – Present

- Architected a three-agent decision-support system on Azure AI Foundry: a Microsoft Fabric data agent over Dataverse, a reasoning agent handling retrieval, calculation and prediction, and a formatting agent for delivery. The system is in active use by managers and VPs, including in board meetings, to support decision-making.
- Equipped the agents with web search, a code interpreter, and function calling into a production gross-reservation forecasting model (145-feature LightGBM; 78.7% directional accuracy vs 70.8% naive-seasonal, 51.7% majority-class), with LLM-as-a-judge evaluation over outputs.
- Built production document-intelligence pipelines processing 7,000–10,000 contracts a month, combining multiple OCR engines with LLM validation, plus a QA review screen that reconciles extracted fields against Dataverse records, flags each mismatch beside its source page, and writes approved corrections back via the Dataverse API, improving data accuracy by 85%.
- Designed and deployed LLM-powered conversational agents for sales assistance and executive reporting on Azure OpenAI and Chainlit, running code-interpreter analysis over CRM, cost and budget data.
- Deployed an autonomous resume-screening agent that parses resumes, scores them against job descriptions, and ranks candidates to automate first-pass HR screening.
- Led and mentored interns building forecasting models, integrating their outputs into production agent recommendations.

Teaching Assistant (Part-time) | Arab Academy for Science & Technology Sept 2024 – Jan 2026

- Taught Data Structures, Introduction to AI, and Advanced AI, covering classical machine learning through artificial neural networks, and mentored students through practical labs on responsible AI, system risk, and data quality.

PROJECTS

- NVIDIA Nemotron Reasoning Agent:** Distilled chain-of-thought traces from a teacher model and QLoRA-finetuned Nemotron-3-Nano-30B (rank ≤ 32) in 4-bit on a single H100; packaged as a vLLM-compatible adapter.
- AMIA Public Challenge 2026 (Kaggle):** Chest X-ray abnormality detection across 14 findings: fused multi-radiologist annotations via weighted box fusion, trained a 5-fold detector ensemble on stratified splits, and gated predictions through a normal/abnormal classifier with multi-scale TTA.
- Motion-S Text-to-Sign Generation:** Built a text-to-sign-language motion model end to end, generating 6 RVQ token layers decoded by a frozen RVQ-VAE, ensembling a MoMask masked transformer with T2M-GPT under classifier-free guidance.
- BirdCLEF+ 2026 (Kaggle):** EfficientNet bird-call classifier on 46k recordings with on-the-fly OGG to log-mel decoding, deployed as INT8 ONNX to score 234 species in under 90 minutes.

EDUCATION

B.Sc. Computer Science | Arab Academy for Science & Technology

Sept 2020 – Jul 2024

GPA 3.8 / 4.0 • Ranked 4th

SKILLS

Agentic AI & Orchestration: Multi-agent systems, function calling & tool use, MCP, LLM-as-a-judge, agent memory & planning, code interpreter & web search, LangChain/LangGraph, Microsoft AutoGen

LLMs & Model Serving: RAG, fine-tuning (LoRA/QLoRA), PEFT/TRL, bitsandbytes, grounding & hallucination guardrails, vLLM, Hugging Face, Azure OpenAI, open models (Llama, Qwen, Gemma-2, Nemotron)

Machine Learning: ML pipelines & model deployment, time-series forecasting, regression & classification, ensemble methods, hyperparameter tuning, feature engineering, stratified k-fold & walk-forward validation, LightGBM, XGBoost

Cloud, Deployment & MLOps: Azure AI Foundry, Azure ML, Azure DevOps, Microsoft Fabric, Dataverse, Docker, REST APIs, Chainlit

Deep Learning & Optimization: PyTorch, TensorFlow, CNNs, Transformers, object detection, TTA & ensembling, OpenCV, INT8/4-bit quantization, ONNX

Programming: Python, C/C++, Java, JavaScript, SQL

CERTIFICATIONS

Microsoft Certified: Azure AI Apps and Agents Developer Associate, 2027 • The Complete Agentic AI Engineering, Udemy • Generative AI Engineering with LLMs, Coursera • Deep Learning Specialization, Coursera